

Harsh Gopinath Chavan

Data Analyst | Data Scientist | Python Developer | SQL | Machine Learning

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PROFESSIONAL SUMMARY

Computer Engineering graduate (Honours in Data Science) skilled in **Python, SQL, machine learning, NLP, and Power BI**, with three end-to-end data projects (two live) and two Python internships.

TECHNICAL SKILLS

Programming: Python, SQL, C++, Java

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, DAX, Microsoft Excel, Exploratory Data Analysis (EDA), Data Cleaning, Data Validation, Data Visualization, KPI Dashboards

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, Natural Language Processing (NLP), NLTK, Generative AI, Google Gemini API, Gradient Boosting, Classification, Feature Engineering, Model Evaluation (ROC-AUC, F1)

Databases: SQL Server, MySQL, MongoDB, T-SQL, Relational Database Design

Web & Backend: Django, Django ORM, REST APIs, Streamlit

Tools & Concepts: Git, GitHub, Jupyter Notebook, OOP, Data Structures, Agile, SDLC

PROJECTS

Predictive Maintenance & Equipment Failure Prediction

[Live Demo](#) | [GitHub](#)

Python, Scikit-learn, Pandas, Gradient Boosting, Streamlit, SQL, Power BI

- Built a **Gradient Boosting** failure classifier on temperature, rotational speed, torque, and tool wear data, reaching **99% accuracy, 96.5% ROC-AUC, and 78% F1 score** on the test set.
- Performed data cleaning, **EDA**, and SQL analysis; built a **Power BI** failure dashboard.
- Deployed a **Streamlit** app on Render that returns failure probability, low/medium/high risk classification, maintenance recommendations, and prediction history.

Workforce Intelligence: HR Attrition Analytics using SQL Server & Power BI

[GitHub](#)

Python, Pandas, SQL Server, T-SQL, Power BI, DAX

- Analyzed **1,470 employee records** using Python (Pandas) EDA and **SQL Server (T-SQL)** scripts for schema creation, data loading, validation, and business analysis.
- Built an interactive **Power BI dashboard** with **DAX KPIs** (total employees, employees left, attrition rate) across tenure, overtime, salary, job satisfaction, department, and job role.
- Found a **16.12% overall attrition rate** (237 of 1,470 employees), highest among 0–1 year tenure (34.88%), low work-life balance (31.25%), and overtime staff (30.53%).

AI Data Analyst: Natural-Language Data Analysis App

[Live Demo](#) | [GitHub](#)

Python, Streamlit, Pandas, Plotly, Google Gemini API, ReportLab

- Built a **Generative AI** app where users upload CSV or Excel files and ask questions in plain English; **Google Gemini** creates an analysis plan that runs locally with **Pandas**.
- Added result validation, **Plotly** charts, AI-generated insights, and PDF report export; tested on retail and HR datasets.
- Cut Gemini API usage by caching analysis plans and running data operations locally; deployed live on Render.

WORK EXPERIENCE

Python Developer Intern, Rubicon Foundation

Jan 2025 – Feb 2025

- Built and maintained **3+ database-driven web apps** with **Python, Django, and MySQL** to automate workflows.
- Designed **normalized relational schemas with 5+ interconnected tables**, preserving data integrity.
- Built secure **REST APIs** with authentication and authorization; optimized Django ORM queries.

Software Developer Intern, Elite Software

Dec 2024 – Feb 2025

- Developed **Python and Django** web application features and resolved production issues.
- Collaborated in **Agile** cycles: sprint planning, code reviews, debugging, testing, and deployment.
- Optimized backend logic and database interactions, improving reliability, maintainability, and response time.
- Enhanced platform usability through backend improvements.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering

2022 – 2026

Savitribai Phule Pune University | *Honours Specialization: Data Science*

CGPA: 7.55/10

PUBLICATION

Automatic Vehicle Speed Control

Jun 2026

International Journal of Scientific Research in Engineering and Management (IJSREM) | [DOI](#)